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JAVA PROJECT REPORT ON

Digital Wallet and Balance Management System

Submitted by

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DECLARATION

I hereby declare that the project entitled "**Digital Wallet and Balance Management System"** submitted in partial fulfillment of the requirements for the course **JAVA PROGRAMMING LANGUAGE** is a **record of my original work** carried out under the guidance of **Dr.BOOPALAN SUNDARAMOORTHY**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

BHARATHKUMAR M

_____

(Your Name)

CERTIFICATE

This is to certify that the project entitled **"Digital Wallet and Balance Management System"** is a **bona fide work** carried out by **BHARATHKUMAR M** , I-'A'(2025-2026),**Department of Computer Science**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **JAVA PROGRAMMING LANGUAGE**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore

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Guide:
(Technical Trainer, KGCAS)

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TABLE OF CONTENTS

S. No.	Topics	Page. No.
1	**Introduction**	
1.1	Objective	
1.2	Overview	
1.3	Features	
2	**Program Analysis**	
2.1	Problem Definition	
2.2	Project Overview	
2.3	Scopes and Limitations	
3	**System Design**	
3.1	Modules	
4	**Methodology**	
4.1	Problem Analysis	
4.2	System Design	
4.3	Program Development	
4.4	Testing and Validation	
4.5	Result and Documentation	
4.6	Algorithm	
4.7	Flowchart	
5	**Source Code**	
6	**Input and Output Screens**	
7	**Discussion and Conclusion**	
7.1	Discussion	
7.2	Conclusion	
8	**References**	

ABSTRACT

The Digital Wallet and Balance Management System is a Java-based application designed to provide a secure and efficient platform for managing digital financial transactions. In traditional systems, users depend on cash or manual record-keeping methods, which can lead to errors, lack of transparency, and difficulty in tracking expenses. This project aims to overcome these issues by introducing a computerized wallet system that maintains accurate records of user transactions.

The system allows users to create an account, add money to their digital wallet, withdraw funds, transfer money, and check their current balance. Each transaction is processed instantly, and the wallet balance is automatically updated. The application ensures that users can perform financial operations quickly and reliably without manual calculations.

The project is developed using Core Java programming concepts such as classes, objects, methods, loops, conditional statements, and exception handling. It can be implemented as a console-based application and can be further enhanced with a graphical user interface using Java Swing.

The main objective of this system is to simplify digital transactions, reduce human errors, and provide a user-friendly interface for managing personal finances. This project also demonstrates the practical implementation of object-oriented programming concepts in developing real-world financial applications.

1. INTRODUCTION

In today's digital era, the use of electronic payment systems has increased rapidly due to the need for fast, secure, and convenient transactions. People are gradually moving away from traditional cash-based systems and adopting digital methods for managing their finances. However, without a proper system, tracking transactions and maintaining accurate balance records can become difficult.

A Digital Wallet and Balance Management System provides a solution to these challenges by offering a platform where users can store money digitally and perform various financial transactions. Instead of relying on physical cash or manual record-keeping, users can manage their funds efficiently through an automated system.

The Digital Wallet system allows users to create an account, add money to their wallet, withdraw funds, transfer money, and check their balance at any time. All transactions are processed instantly, and the system automatically updates the wallet balance after each operation. This reduces the chances of errors and improves accuracy.

This project is developed using Core Java programming concepts such as classes, objects, methods, loops, and conditional statements. The application can be implemented as a console-based program or further enhanced using Java Swing for a graphical user interface.

The main goal of this system is to provide a simple, secure, and efficient way to manage digital transactions. It also helps in understanding how Java can be used to develop real-world financial applications.

1.1 Objective

The main objective of the Digital Wallet and Balance Management System is to develop a simple and efficient application using Core Java that enables users to manage their digital financial transactions easily and securely.

The specific objectives of this project are:

- To design and develop a digital wallet system using Java programming.
- To allow users to create and manage their wallet accounts.
- To enable users to add money (deposit) into their wallet.
- To provide functionality for withdrawing money from the wallet.
- To allow users to transfer money between accounts.
- To automatically update and display the current wallet balance.
- To reduce errors associated with manual financial calculations.
- To provide a user-friendly interface for performing transactions.
- To demonstrate the implementation of Core Java concepts such as classes, objects, methods, loops, and conditional statements.

1.2 Overview

The Digital Wallet and Balance Management System is a menu-driven application developed using the Java programming language. It provides a digital platform where users can manage their financial transactions in a simple and efficient manner.

The system allows users to create a wallet account and perform various operations such as adding money, withdrawing funds, transferring money, and checking their current balance. Each transaction is processed instantly, and the system automatically updates the wallet balance after every operation.

The application displays a set of options in the form of a menu, enabling users to choose the desired operation. Based on the user's input, the system performs the corresponding action and provides immediate feedback. This helps in reducing manual effort and ensures accurate financial calculations.

The system is designed to be user-friendly and easy to operate, making it suitable for beginners as well as small-scale applications. It can be implemented as a console-based program and can also be enhanced with a graphical user interface using Java Swing for better usability.

Overall, this project demonstrates how Core Java concepts can be applied to develop a real-world application for managing digital transactions and maintaining wallet balances efficiently.

1.3 Features

The Digital Wallet and Balance Management System includes the following key features:

- **Account Creation** – Allows users to create a new digital wallet account with basic details.
- **Add Money (Deposit)** – Enables users to add money to their wallet easily, increasing their available balance.
- **Withdraw Money** – Allows users to withdraw money from their wallet based on the available balance.
- **Money Transfer** – Provides the functionality to send money from one wallet to another.
- **Balance Check** – Displays the current wallet balance instantly after each transaction.
- **Transaction Processing** – Performs all financial operations automatically and updates the balance in real time.
- **Error Handling** – Handles invalid inputs and prevents transactions like withdrawing more than the available balance.

- **User-Friendly Interface** – Offers a simple menu-driven interface that is easy to understand and operate.
- **Fast and Accurate Operations** – Ensures quick processing of transactions with accurate calculations.
- **Scalability (Future Enhancement)** – Can be extended to include features like transaction history, PIN security, database integration, and mobile app support.

2. PROGRAM ANALYSIS

Program Analysis is an important phase in software development where the problem is studied in detail and system requirements are clearly identified. This stage helps in understanding how the system should function and what features are necessary to solve the problem effectively.

In the case of the Digital Wallet and Balance Management System, the analysis focuses on identifying the limitations of traditional money management methods and designing a computerized solution to overcome those issues.

In many situations, people rely on cash transactions or manual tracking of expenses, which can lead to several problems such as calculation errors, lack of proper records, and difficulty in managing finances efficiently. There is also a risk of losing money or failing to track spending accurately.

To address these issues, the Digital Wallet System is designed to provide a secure and automated platform for managing financial transactions. The system ensures that all operations such as adding money, withdrawing funds, and transferring money are handled efficiently and accurately.

The analysis phase also helps in defining system requirements such as user account management, transaction processing, balance updates, and error handling. By understanding these requirements, the system can be designed in a way that is both user-friendly and efficient.

Overall, Program Analysis ensures that the system is well-planned and capable of solving real-world financial management problems effectively.

2.1 Problem Definition

In many traditional financial systems, people rely on physical cash or manual methods to manage their money. These methods often create several challenges in handling financial transactions efficiently and accurately.

Some common problems include:

- **Lack of Proper Tracking** – It is difficult to keep accurate records of income, expenses, and transactions manually.
- **Calculation Errors** – Manual calculations can lead to mistakes in balance management.
- **Time-Consuming Process** – Recording and verifying transactions manually takes more time.
- **Risk of Loss or Theft** – Physical cash can be lost or stolen easily.
- **No Real-Time Updates** – Users cannot instantly know their current balance after transactions.
- **Difficulty in Managing Multiple Transactions** – Handling multiple payments and transfers becomes complicated.

Due to these issues, there is a need for a digital system that can manage financial transactions efficiently and securely.

The Digital Wallet and Balance Management System is designed to overcome these problems by providing an automated solution that maintains accurate records, performs instant calculations, and updates the wallet balance in real time.

2.2 Project Overview

The Digital Wallet and Balance Management System is a Java-based application developed to simplify and manage digital financial transactions efficiently. The system provides a platform where users can store money digitally and perform various operations such as adding money, withdrawing funds, transferring money, and checking their balance.

The application follows a menu-driven approach, allowing users to interact with the system by selecting options based on their requirements. Once a user performs any transaction, the system automatically processes the request and updates the wallet balance in real time.

The system includes the following functionalities:

- Creating and managing a digital wallet account
- Adding money (deposit) to the wallet
- Withdrawing money from the wallet
- Transferring money between wallets
- Checking the current wallet balance
- Handling invalid inputs and insufficient balance conditions

This project is implemented using Core Java programming concepts such as classes, objects, methods, loops, and conditional statements. It can be developed as a console-based application and can also be enhanced with a graphical user interface using Java Swing.

Overall, the system provides a simple, fast, and reliable way to manage digital transactions and demonstrates the practical application of Java programming in real-world financial systems.

2.3 Scopes and Limitations

Scope

The scope of the Digital Wallet and Balance Management System defines the range of functionalities and areas where the system can be effectively used. The system is designed to provide basic digital financial management features and can be applied in the following areas:

- **Personal Finance Management – Helps users manage their daily transactions and track wallet balance.**
- **Small-Scale Payment Systems – Suitable for small businesses or local transactions.**

- **Educational Purpose – Useful for students to understand real-world application of Java programming concepts.**
- **Cashless Transactions – Encourages digital payments instead of physical cash usage.**
- **Basic Transaction Handling – Supports deposit, withdrawal, and fund transfer operations.**

Limitations

While the system provides essential features, it also has certain limitations:

- **No Real Bank Integration – The system does not connect with actual banking systems or payment gateways.**
- **Limited Security Features – Basic version may not include advanced security like encryption or authentication.**
- **Single User System – Typically designed for one user unless extended.**
- **No Database Storage – Data may not be stored permanently without database integration.**
- **Limited Scalability – Not suitable for large-scale financial systems without further development.**

Overall, the system is ideal for learning and small-scale applications but can be enhanced in the future by adding features such as database connectivity, user authentication, and advanced security mechanisms.

3. SYSTEM DESIGN

System Design is the phase where the overall structure of the system is planned before implementation. In this stage, the system is divided into different modules, and the workflow is defined clearly. A well-designed system helps in developing the application efficiently and makes it easy to understand, maintain, and upgrade.

The Digital Wallet and Balance Management System is designed as a menu-driven application where users can perform various financial operations such as adding money, withdrawing funds, transferring money, and checking their balance.

The system is divided into multiple modules, each responsible for a specific function.

3.1 Modules

The main modules of the Digital Wallet and Balance Management System are:

---

1. User Account Module

This module is responsible for managing user account details.

Functions:

- Create a new wallet account
- Store user information (name, ID, etc.)
- Initialize wallet balance

---

2. Wallet Management Module

This module handles the wallet balance.

Functions:

- Maintain current balance
- Update balance after each transaction
- Ensure balance consistency

---

3. Transaction Module

This module manages all financial transactions.

Functions:

- Add money (deposit)

- Withdraw money
- Transfer money
- Validate transaction conditions (e.g., sufficient balance)

---

4. Balance Inquiry Module

This module allows users to check their current balance.

Functions:

- Display updated wallet balance
- Provide instant balance information

---

5. Error Handling Module

This module ensures smooth system operation.

Functions:

- Handle invalid inputs
- Prevent overdraft (insufficient balance)
- Display error messages

---

6. Exit Module

This module allows the user to safely exit the application.

Functions:

- Terminate the program
- Ensure proper closure of the system

4. METHODOLOGY

Methodology describes the step-by-step process followed to develop the Digital Wallet and Balance Management System. It explains how the problem is analyzed, how the system is designed, implemented, tested, and documented. A structured approach ensures that the system works efficiently and meets all requirements.

The development of this project follows these main stages:

- Problem Analysis
- System Design
- Program Development
- Testing and Validation
- Result and Documentation

4.1 Problem Analysis

In this phase, the existing issues in traditional money management systems are studied. Problems such as manual errors, lack of proper tracking, and difficulty in handling transactions are identified.

Based on these issues, the need for a digital wallet system is understood, and requirements such as account management, transaction handling, and balance updates are defined.

4.2 System Design

In this stage, the structure of the system is planned. The system is divided into modules such as:

- User Account Module
- Wallet Management Module
- Transaction Module
- Balance Inquiry Module
- Error Handling Module

The flow of the program is designed so that users can easily interact with the system through a menu-driven interface.

4.3 Program Development

In this phase, the actual coding of the system is done using Java programming language.

The following steps are involved:

- Creating classes and objects
- Writing methods for each function
- Implementing transaction logic
- Handling user inputs
- Updating wallet balance

Core Java concepts such as loops, conditional statements, and exception handling are used.

4.4 Testing and Validation

This phase ensures that the system works correctly and produces accurate results.

Different test cases are used to verify system functionality.

Example Test Cases:

Test Case	Expected Result
Add money	Balance increases correctly
Withdraw money	Balance decreases correctly
Transfer money	Amount transferred successfully
Invalid input	Error message displayed

This step helps in identifying and fixing errors.

4.5 Result and Documentation

After successful testing, the results of the system are recorded and documented.

- The system performs all wallet operations correctly
- Balance updates are accurate
- Outputs are displayed clearly

Documentation includes explanation of system design, code, and outputs.

4.6 Algorithm

Algorithm for Digital Wallet System:

1. Start the program
2. Create user account
3. Display menu options
4. Select operation (Deposit / Withdraw / Transfer / Check Balance)
5. Enter required details

6. Perform transaction
7. Update wallet balance
8. Display result
9. Repeat until user exits
10. End program

4.7 Flowchart

Students should draw a flowchart representing the program flow.

Basic Flow:

Start
↓
Display Menu
↓
Select Option
↓
Perform Operation
↓
Update Balance
↓
Display Result
↓
End

You can draw the flowchart using tools like:

- draw.io
- Lucidchart
- MS Vision

5. SOURCE CODE

```
import java.util.Scanner;

class DigitalWallet {
    private String userName;
    private double balance;

    // Constructor
    DigitalWallet(String name) {
        this.userName = name;
        this.balance = 0.0;
    }

    // Deposit Money
    public void deposit(double amount) {
        if (amount > 0) {
            balance += amount;
            System.out.println("Amount Deposited Successfully!");
        } else {
            System.out.println("Invalid Amount!");
        }
    }

    // Withdraw Money
    public void withdraw(double amount) {
```

```
        if (amount <= balance && amount > 0) {
            balance -= amount;
            System.out.println("Amount Withdrawn Successfully!");
        } else {
            System.out.println("Insufficient Balance or Invalid Amount!");
        }
    }

    // Transfer Money (simulation)
    public void transfer(double amount) {
        if (amount <= balance && amount > 0) {
            balance -= amount;
            System.out.println("Money Transferred Successfully!");
        } else {
            System.out.println("Transaction Failed!");
        }
    }

    // Check Balance
    public void checkBalance() {
        System.out.println("Current Balance: ₹" + balance);
    }

    // Display User
    public void displayUser() {
        System.out.println("User: " + userName);
    }
}
```

```
public class WalletApp {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter your name: ");
        String name = sc.nextLine();

        DigitalWallet wallet = new DigitalWallet(name);

        int choice;

        do {
            System.out.println("\n--- DIGITAL WALLET MENU ---");
            System.out.println("1. Deposit Money");
            System.out.println("2. Withdraw Money");
            System.out.println("3. Transfer Money");
            System.out.println("4. Check Balance");
            System.out.println("5. Exit");

            System.out.print("Enter your choice: ");
            choice = sc.nextInt();

            switch (choice) {
                case 1:
                    System.out.print("Enter amount to deposit: ");
                    double deposit = sc.nextDouble();
                    wallet.deposit(deposit);
                    break;
```

```
            case 2:
                System.out.print("Enter amount to withdraw: ");
                double withdraw = sc.nextDouble();
                wallet.withdraw(withdraw);
                break;

            case 3:
                System.out.print("Enter amount to transfer: ");
                double transfer = sc.nextDouble();
                wallet.transfer(transfer);
                break;

            case 4:
                wallet.checkBalance();
                break;

            case 5:
                System.out.println("Thank you for using Digital Wallet!");
                break;

            default:
                System.out.println("Invalid Choice!");
        }

    } while (choice != 5);

    sc.close();
  }
}
```

6. INPUT AND OUTPUT SCREENS

User Login / Name Input Screen

Input:

Enter your name: Arun

Output:

Welcome, Bharathkumar!

Main Menu Screen

--- DIGITAL WALLET MENU ---

1. Deposit Money

2. Withdraw Money

3. Transfer Money

4. Check Balance

5. Exit

Figure 1: Deposit Money Screen

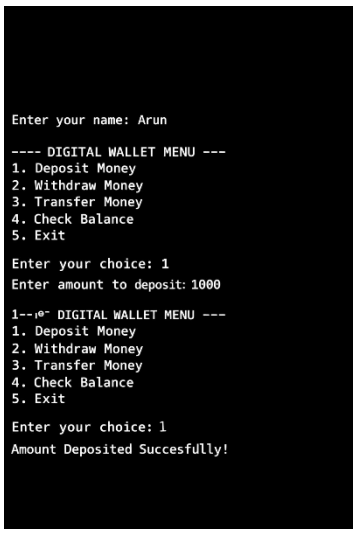


Figure 2: Withdraw Money Screen

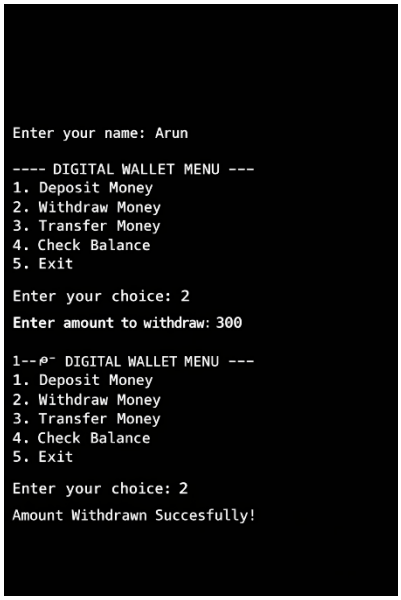


Figure 3: Transfer Money Screen

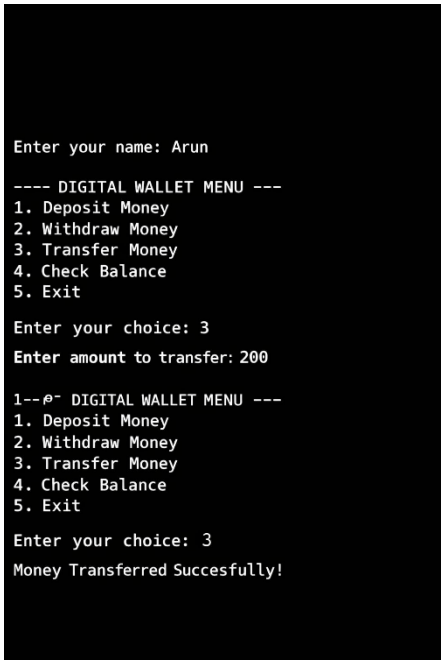


Figure 4: Balance Check Screen

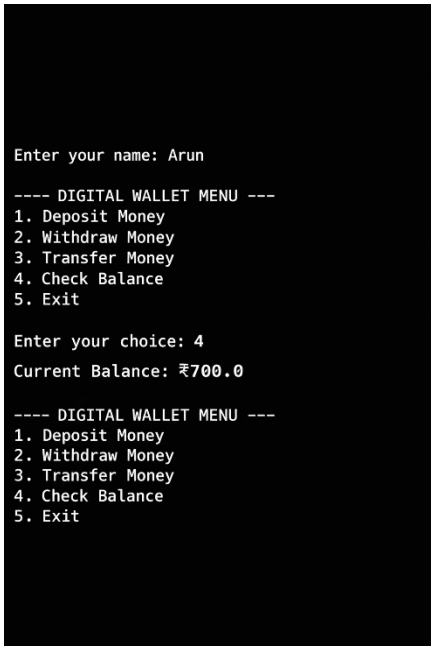
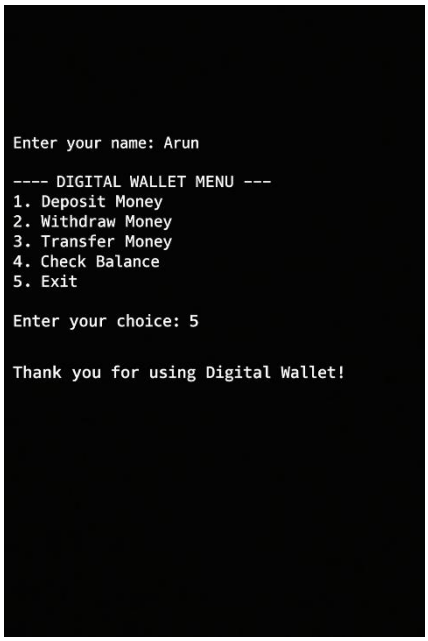


Figure 5: Exit Screen



7. DISCUSSION & CONCLUSION

7.1 Discussion

The Digital Wallet and Balance Management System was developed to provide a simple and efficient way of handling digital financial transactions. The system allows users to perform essential operations such as depositing money, withdrawing funds, transferring money, and checking their wallet balance.

During the development of this project, Core Java concepts such as classes, objects, methods, loops, and conditional statements were effectively used. The system was designed as a menu-driven application, making it easy for users to interact and perform operations without complexity.

The application was tested with different inputs to ensure that all functionalities work correctly. The results show that the system performs accurate calculations and updates the wallet balance instantly after each transaction. Error handling mechanisms were also implemented to manage invalid inputs and insufficient balance conditions.

Overall, the system reduces manual effort, minimizes calculation errors, and provides a faster and more reliable way to manage financial transactions.

7.2 Conclusion

The Digital Wallet and Balance Management System successfully demonstrates how Java programming can be used to develop real-world financial applications. The system provides a user-friendly platform for managing digital transactions and maintaining wallet balance efficiently.

By automating financial operations, the system reduces errors, improves accuracy, and saves time compared to traditional methods. It also helps users understand the importance of digital payment systems in modern society.

This project serves as a strong foundation for developing more advanced financial systems. In the future, it can be enhanced by adding features such as user authentication, database integration, transaction history, and graphical user interfaces.

In conclusion, the project achieves its objectives and proves to be a useful application for learning and implementing Core Java concepts in practical scenarios.

8. REFERENCES

The following resources were referred to during the development of the Digital Wallet and Balance Management System:

Books

- Herbert Schildt – *Java: The Complete Reference*
- E. Balagurusamy – *Programming with Java*

Websites

- https://www.oracle.com/java
- https://www.geeksforgeeks.org
- https://www.w3schools.com
- https://www.javatpoint.com